



User manual for data and metadata transmission: EnvLogger

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1. Scope of the document

This manual provides step-by-step instructions for transmitting data collected by the EnvLogger (<https://www.scoop-ocean.org/catalogue/TOOL2223/envlogger>) temperature sensor and related metadata. It is intended to guide users through all necessary operations regarding downloading, sharing, and sending the recorded data to the data collection system. Following this guide ensures accurate data and metadata handling and helps maximise the operational life of the EnvLogger sensor.

EnvLogger is a small temperature sensor made waterproof through 3D-printed encapsulation. These sensors are equipped with an NFC proximity sensor for interaction with external devices.

The sensor is fully programmable via a dedicated mobile application, which allows users to select the sampling frequency, download data, and start the device. The sensor is fitted with a non-rechargeable internal battery and limited memory capacity; its lifespan varies depending on the sampling frequency selected. For continuous use, a temperature sampling interval of every 10 minutes while in motion is recommended. This maximises operational capacity, ensuring approximately 365 days of data collection. Each time the logger connects to a mobile device, it provides details of the maximum possible sampling duration.

For further details, please refer to the manufacturer's website:

<https://electricblue.eu/temperature-envloggers>.

2. Data collection

To interact with the sensors, it is necessary to have a smartphone equipped with an NFC sensor and with the EnvLogger Viewer application installed, which is available for both iOS and Android.

For a guide on using the application, we recommend consulting the user manual provided by the manufacturer: <https://electricblue.eu/temperature-envloggers>.

In brief, the steps to be carried out are:

1. Start the loggers;
2. Submerge the sensor to record data;
3. Download the data to the device once the session is complete.

2.1. Sensor serial number

Each device is assigned a unique serial number that allows it to be unambiguously identified. This information is essential at different stages of the process, both for data management and for metadata definition.

When you connect the sensor to your mobile phone (see the user manual provided by the manufacturer: <https://electricblue.eu/temperature-envloggers>), it is possible to identify the serial number of the specific device, as shown in Figure 1.

When reporting the sensor serial number, please preserve the spaces within the sensor serial number (e.g. 0489 CD00 B81D 01).



Figure 1. Sensor serial number

2.2. Spatial information about the sampling

While sampling, make a note of the coordinates of the location(s) where you are sampling and the depth(s) at which you are doing so. See details in “0 Location and depth”.

2.3. Data download

Once sampling is complete, to download the data to your device, simply connect the logger to your phone and click on “DOWNLOAD DATA” (Figure 2).

IMPORTANT:

- **Do not click the “STOP” button, as this will erase the data;**
- **Wait for the download to finish and do not move the phone or the sensor during the process, to avoid interrupting the connection;**
- If the warning message “Hold on, data are still being downloaded...” appears, click “OK” and allow the operation to complete. The process may take a considerable amount of time.

Once the download is complete, a screen showing the latest mission will appear, displaying the graph of the data collected during the mission (Figure 3).



Figure 2. Data download



Figure 3. Data of the last mission

2.4. Sending data to the data sharing system

The data can be transmitted through two distinct channels. The first option is via email, sent directly from the EnvLogger application installed on the smartphone used for data



collection. This method enables immediate sharing of the recorded information, allowing users to quickly export and forward datasets as email attachments to designated recipients. Alternatively, the data may be transferred through an institutional data server of the partner organisation responsible for data collection.

2.4.1. Sending data to the data sharing system via email

Through the EnvLogger application installed on a smartphone, users can share collected data either immediately at the end of an individual measurement session or at a later time, offering flexibility for situations in which immediate transmission is not practical.

Note that the procedure differs slightly depending on when you share the data: if you share immediately after downloading, the mission must remain active until the data has been sent; if you share at a later time, the mission must be stopped first before accessing the data folder.

Sending data after the collection

It is possible to share the file directly at the end of a single sampling session, immediately after downloading it.

Important: do not click the “STOP” button.

Make sure to keep the sensor connected to the phone at all times, then click “SHARE” from the latest mission screen (Figure 3).

On the sharing screen, select your email service and send the automatically generated email to the data collection address: landsealot@s4oceandata.eu (Figure 4).

IMPORTANT:

- **Do not change the name of the files** as it contains information relating to the specific sensor (sensor serial number);
- **Do not change the subject of the email:** it is automatically populated with the sensor type (Envlogger) and serial number;
- There is no need to add a body to the email.

After sending, the mission must be stopped by clicking the “STOP” button on the latest mission screen (Figure 3).

In this way the sensor stops collecting data. To start a new measurement, repeat all the steps as indicated in the user manual provided by the manufacturer: <https://electricblue.eu/temperature-envloggers>.



Figure 4. Sending data via email

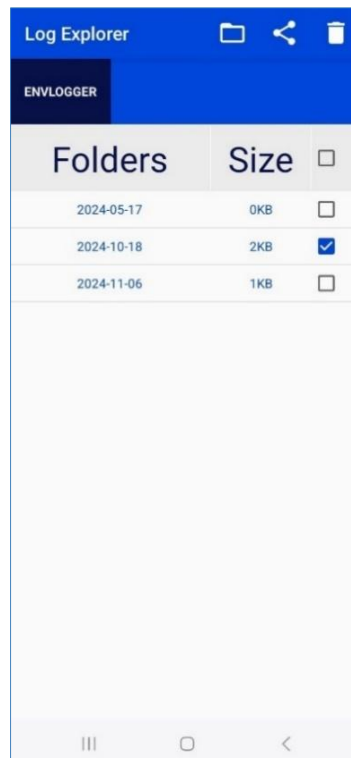


Figure 5. Selection and sharing of folders

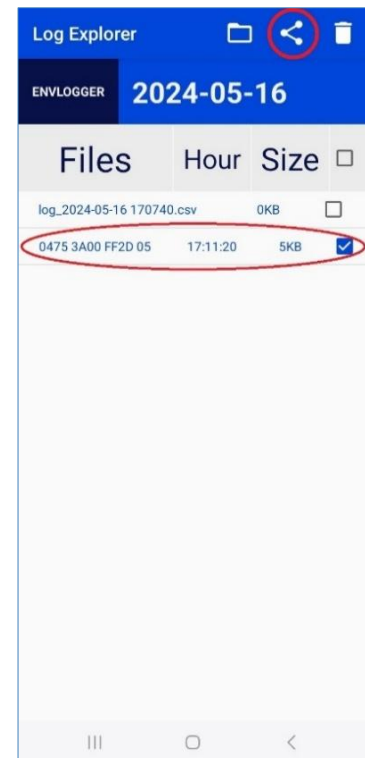


Figure 6. Selection and sharing of files

Sending data at a later time

In this case, to complete the mission, it is necessary to stop the mission by clicking the “STOP” button on the latest mission screen (Figure 3).

From the app’s home screen, access the data folder by clicking the icon at the top right of the page (Figure 3).

The data are organised into subfolders divided by day. You can proceed in two different ways:

- Share one or more day folders: select them that you wish to share and press the “SHARE” button (Figure 5).
- Share single files: open the folder for the current day to view all CSV files containing the recorded data. Select the file(s) named with the sensor’s serial number (not the file named “log_...”) that you wish to share and press the “SHARE” button (Figure 6).

On the sharing screen, select your email service and send the automatically generated email to the data collection address: landsealot@s4oceandata.eu (Figure 4).

To start a new measurement, repeat all the steps as indicated in the user manual provided by the manufacturer: <https://electricblue.eu/temperature-envloggers>.

IMPORTANT:

- **Do not change the name of the files** as it contains information relating to the specific sensor (sensor serial number);
- **Do not change the subject of the email:** it is automatically populated with the sensor type (Envlogger) and serial number;



- There is no need to add a body to the email.

2.4.2. Sending data to the data sharing system via data server

If the partner responsible for collecting the data prefers to transfer it to their own data server (e.g. ERDAPP™), it is possible to establish a direct connection between that server and the data sharing service of LandSeLot.

To enable this arrangement, the partner must contact ETT in order to jointly define the technical and operational procedures required for integration.

This process typically involves agreeing on data formats, transmission protocols, security standards, and authentication mechanisms to ensure full interoperability between systems. By setting up a dedicated connection, partners can automate data flows, maintain control over their internal infrastructure, and guarantee secure, reliable synchronisation with the LandSeLot platform.

3. Data geolocation

Data can be collected with EnvLogger in two different modes: 1) at a fixed location; 2) along a trajectory.

3.1. Fixed-location measurements

When collecting data at a fixed point, the geographical position must be specified using the dedicated metadata entry form within the application. Accurate completion of the metadata is essential to ensure proper georeferencing and subsequent data validation (see chapter “4 Metadata”).

3.2. Trajectory-based measurements

When collecting data along a trajectory, a GPS device is required. The GPS must record positional data in GPX format, which will later be used to associate each measurement with its corresponding geographic coordinates.

Ensure that both the sensor and the GPS device are configured to the same time zone. This is essential to correctly match the recorded measurements with the corresponding positional data during post-processing. To check or change the time zone on your GPS, go to the device settings and look for a 'Time' or 'System' section. If your GPS does not have a time zone setting, ensure that it is set to record in UTC and note this for post-processing.

Procedure for trajectory-based data collection:

1. Before deploying the sensor in the water, activate the GPS device;
2. At the end of the mission, save the GPX file generated by the GPS;
3. Rename the file with the following format:

Envlogger + sensor serial number
(example: Envlogger 0489 CD00 B81D 01)

Please note carefully – file name:

- Write “Envlogger” exactly as shown (not “EnvLogger”, “envlogger”, etc.).
- Preserve the spaces within the sensor serial number (e.g. 0489 CD00 B81D 01) (see chapter “2.1 Sensor serial number”).

Failure to comply with these formatting requirements may prevent the correct association of the GPX file with the corresponding sensor data.

4. Send the GPX file in one of the following ways:
 - a. Upload directly in the metadata form (see chapter “4 Metadata - Location and depth”);
 - b. Send to the data collection email address: landsealot@s4oceandata.eu writing the subject of the email with the following format:

Envlogger + sensor serial number
(example: Envlogger 0489 CD00 B81D 01)



Please note carefully – email subject:

- Write “Envlogger” exactly as shown (not “EnvLogger”, “envlogger”, etc.).
- Preserve the spaces within the sensor serial number (e.g. 0489 CD00 B81D 01) (see chapter “2.1 Sensor serial number”).

Failure to comply with these formatting requirements may prevent the correct association of the GPX file with the corresponding sensor data.



4. Metadata

This section is designed to help users, even those without technical or scientific experience, to correctly provide information describing how and when the data was collected. Each time a measurement is carried out, it is important to complete a dedicated form.

A **measurement** is defined as a data collection activity that:

- is carried out using the same instrument (device),
- takes place at a specific point in time,
- is performed by the same person or by the same group of people.

Information that helps to describe, identify, classify and contextualise environmental data collected through measurements is known as **metadata**. This includes aspects such as provenance, methods of acquisition, geographical location, and the device used.

Completing the form is essential, even if it may seem like a formal step. The information you provide helps researchers, regulatory bodies, and decision-making systems to better understand the data, and to ensure its quality, reliability, traceability, and comparability. This manual will guide you through the process step by step, explaining what each field requires and why it matters.

Therefore, it is important for us that you **help us** describe the data you have collected, and we kindly ask you to complete the form available at the following link:

<https://ec.europa.eu/eusurvey/runner/sensor-deployment>

The metadata form is divided into 4 sections:

1. General information about the activity
2. Information about the user
3. Description of the device
4. Description of the measurement

4.1. General information about the activity

You must first indicate the **project** to which the activities belong: please select “Land-Sea interface: Let’s observe together! (LandSeaLot)”.

After that, you will be asked for specific information that must be provided by the activity leader: the name of the specific **LandSeaLot Integration Lab** where you are located and the organisation responsible for the activity.

4.2. Information about the user

First, you must state whether you are a **scientist** (who is part of the LandSeaLot project) or a **citizen scientist**.

If you are a **citizen scientist**, you will be asked whether you agree to have your name (and of your collaborators) associated with the data you collected, or whether you prefer to



remain anonymous. This matters because the data you collect may be used in scientific publications, reports, or public databases and if you agree, your name will appear as a contributor, giving you formal recognition for your work. If you prefer to remain anonymous, your data will still be used, but your name will not appear anywhere.

If you agree to be named, you will be asked for your name and the names of any other people who helped collect the data (if they also agree to be named).

You consent to the processing of your personal data (first name and surname) in accordance with Regulation (EU) 2016/679 (GDPR), solely for project-related purposes linked to the collection of environmental data and their sharing with the scientific community through open access repositories for research and publication purposes.

If you are a scientist, you will be asked to provide information about yourself and your collaborators.

In both cases, please provide names in the following format: **Surname, Name** (e.g., Smith, John).

You will also be asked to indicate the **type of creator**: Person, Group, or Institution.

Optionally, you can provide your **ORCID** — a unique identifier used in the scientific community to unambiguously associate researchers with their work. If you have one, you can find it at <https://orcid.org>. If you do not have one, you can leave this field blank.

4.3. Description of the device

You must select the **type of sensor** you used (“EnvLogger”), indicate its **serial number**, and specify the **start date and time of data collection** (i.e., the date and time on which you began recording data with the device).

For the sensor serial number, it is important to follow the instructions reported in the chapter “2.1 Sensor serial number”.

4.4. Description of the measurement

This is the most substantial but essential section to describe in detail what you have done.

General description

You are asked to describe the **activity** carried out, covering the following aspects: **what** data was collected, **how** it was collected, **where** the measurement took place, and **who** was involved. For example: “*The sampling took place in Loco Beach as part of a citizen science activity. The sensor was mounted on a rock face to continuously collect data at that specific location for 10 hours...*”

Keywords

You must then provide some **keywords** and/or tags - words or a short phrase that clearly represent the main topic of a piece of content. They act as labels that help describe what a data refers to, classify it, and enable others to find it more easily. For example, if you have data on water quality, some keywords might be: water, salinity, monitoring, ocean.



Location and depth

First, you will be asked whether the device was deployed at a **fixed location** or **along a trajectory**. Information about the **location** of the measurement is then required:

- If it was carried out at a fixed point, you must provide the geographical coordinates (latitude and longitude) in the WGS84 format. You can obtain this from a GPS device or, more simply, copy them from Google Maps: right-click on the desired point on the map and select the coordinates that appear at the top of the menu — they will be automatically copied to your clipboard in the correct format (e.g. 44.12345, 8.98765).;
- If the measurement was taken at multiple points or along a trajectory, you can provide the information in two ways:
 - If you have a GPS device, you can provide the GPX file as described in section 3.2 Trajectory-based measurements,
 - If you do not have a GPS device, manually enter the working area by indicating the maximum and minimum latitude and longitude values in WGS84 format, which you can simply copy from Google Maps using the same method described above.

You must then indicate whether the measurements were taken at **one fixed depth** or at **different depths** and enter the depth value in meters.

Platform

A platform is any physical or technological system that supports and enables the collection of measurements in a given environment, such as vessels, fixed stations, structures, or sampling points.

If applicable, you may also enter a **pre-existing platform identifier** (e.g. an official name or code assigned to the vessel or station) and any **additional information on the deployment** that may be useful for interpreting the data.

The list of possible platforms:

- beach/intertidal zone structure: A structure to which instrumentation may be attached that is either in air or under water depending on the state of the tide and weather conditions.
- coastal structure: A fixed man-made structure permanently linked to land with access to water at all states of the tide to which instrumentation may be attached (e.g. pier).
- fishing vessel: A platform operating on the surface of the water column whose primary purpose is the commercial harvesting of fish or shellfish but may be engaged in scientific activities such as fish stock surveys or mooring deployments and recoveries.
- fixed subsurface vertical profiler: A platform that periodically makes an automated vertical traverse of the water column at a predetermined fixed location. (e.g. YSI vertical profiler, HOMER CTD).
- float: A free-floating platform either on the surface of the water column or at a predetermined depth within the water column.
- human: A human being without specialised equipment operating on land or the surface of the water column.
- land or seafloor: A platform located on the solid surface of the Earth either above or below sea level.

- land/onshore structure: A fixed man-made structure on land to which instrumentation may be attached (e.g. meteorological tower).
- man-powered boat: A platform operating on the surface of the water column that is manually propelled and may not be easily removed from the water (e.g. trireme).
- moored surface buoy: An unmanned instrumented platform operating on the surface of the water column loosely tethered to the seafloor to maintain a fixed position (e.g. ODAS buoy).
- mooring: A tethered collection of oceanographic instruments at a fixed location that may include seafloor, mid-water and surface components.
- naval vessel: A platform operating on the surface of the water column in unpredictable locations that is primarily equipped, manned and operated for military purposes. Includes surface warships of all sizes and logistic support vessels.
- offshore structure: A fixed (for the duration of the measurements) man-made structure away from the coast to which instrumentation may be attached (e.g. oil rig, gas rig or jack-up barge).
- research vessel: A platform of any size operating on the surface of the water column in unpredictable locations that is specifically equipped, manned and operated for scientific, usually oceanographic, research.
- river station: An instrumented structure in a river upstream of its tidal limit.
- self-propelled boat: A small self-propelled platform operating on the surface of the water column in unpredictable locations that is smaller than a ship, but too large to easily remove from the water.
- ship: A large platform operating on the surface of the water column. Objective definitions for guidelines: >50m length (EU), >100 foot length (USA), >300 GRT weight (SOLAS). Subjective definition: a ship is a vessel big enough to carry a boat.
- subsurface mooring: A collection of oceanographic instruments attached to wires suspended between anchors on the seabed and buoyant spheres in the water column.
- surface vessel: A mobile platform with propulsion operating on and restricted to the surface of a water body.
- vessel at fixed position: A platform of any size occupying a fixed location on the surface of the water column for prolonged periods collecting scientific (oceanographic and meteorological) data as a primary or secondary mission. Includes light vessels and weather ships.
- vessel of opportunity: A platform for purpose of commerce of any size operating on the surface of the water column in unpredictable locations that regularly collects scientific (oceanographic and meteorological) data (e.g. an instrumented cargo vessel).
- vessel of opportunity on fixed route: A platform repeatedly following a predictable fixed track on the surface of the water column that collects scientific (oceanographic and meteorological) data (e.g. an instrumented ferry).
- Other: any other platform not indicated in the list according to the NERC L06 Vocabulary, <https://vocab.nerc.ac.uk/collection/L06/current/>.

Some of “Other”:

- autonomous surface water vehicle: A self-propelled vehicle operating on the sea surface with no human occupants.



- autonomous underwater vehicle: A free-roving platform operating in the water column with propulsion but no human operator on board (e.g. Autosub, Gavia).
- diver: A human being with self-contained equipment or surface-connected suit enabling operation within the water column.
- drifting manned submersible: A platform operating in the water column attached to a mothership by an umbilical but with no means of propulsion that has at least one human operator on board (e.g. bathysphere).
- drifting subsurface float: An unmanned instrumented platform drifting freely in the water column at a depth governed by its density (e.g. Swallow float).
- drifting subsurface profiling float: An unmanned instrumented platform drifting freely in the water column that periodically makes vertical traverses through the water column (e.g. Argo float).
- drifting surface float: An unmanned instrumented platform operating on the surface of the water column often attached to a drogue to track currents rather than winds (e.g. Argos buoy).
- fixed benthic node: A collection of oceanographic instruments mounted at a fixed position on the seabed (e.g. POL Monitoring Platform, seabed ADCP).
- ice island: A floating ice sheet detached from the coast.
- ice shelf: A floating ice sheet attached to the coast.
- Ice-tethered subsurface profiling float: A platform that periodically makes an automated vertical traverse of the water column that is fixed to a floating body of ice.
- kite: An aerofoil tethered to the ground held aloft by the wind.
- lowered unmanned submersible: An unmanned platform lowered and raised vertically by a cable from the mothership. Includes any type of profiling sensor mounting such as CTD frames, profiling radiometers and instrumented nets.
- man-powered small boat: A platform operating on the surface of the water column that is manually propelled and may be easily removed from the water (e.g. rowing boat, canoe).
- propelled manned submersible: A platform operating in the water column that has both self-contained propulsion and at least one human operator on board (e.g. submarine).
- propelled unmanned submersible: A platform operating in the water column attached to a mothership by an umbilical with limited propulsion and no human operator on board (e.g. ROV).
- self-propelled small boat: A small self-propelled platform operating on the surface of the water column that may be easily removed from the water (e.g. shore-based RIBs, ships' boats).
- sub-surface gliders: Platforms with buoyancy-based propulsion that are capable of operations at variable depths which are not constrained to be near the sea surface.
- submersible: A platform operating within a water body.
- surface gliders: Platforms operating at a single depth near the sea surface, using a combination of solar energy and wave motion as means of propulsion.